

POWER BI

Basic Questions & Answers

LEVEL	TOPICS	USE
Beginner	Power BI, Power Query, Data Model, DAX, Visuals	Interview • Viva • Practice

1. Power BI Fundamentals

1. What is Power BI?

Answer: Power BI is a Microsoft business intelligence and data visualization tool used to connect, transform, analyze, and visualize data.

2. Why is Power BI used?

Answer: It helps users turn raw data into interactive reports and dashboards for better data analysis and decision-making.

3. What are the main components of Power BI?

Answer: Power BI Desktop, Power BI Service, Power BI Mobile, Power Query, and the Power BI Report Server.

4. What is Power BI Desktop?

Answer: A Windows application used to connect data, transform it, create data models, build visuals, and design reports.

5. What is Power BI Service?

Answer: A cloud-based platform where reports and dashboards can be published, shared, refreshed, and managed.

6. What is a dashboard?

Answer: A single-page collection of visuals that gives a quick overview of important information.

7. What is a report?

Answer: A Power BI file containing one or more pages of interactive visualizations based on a data model.

2. Data & Power Query

8. What is Power Query?

Answer: Power Query is the data preparation and transformation engine used to clean and reshape data before analysis.

9. What is ETL?

Answer: ETL means Extract, Transform, Load: obtaining data, cleaning/transforming it, and loading it for analysis.

10. What is data cleaning?

Answer: The process of fixing missing values, duplicates, incorrect data types, inconsistent text, and other data-quality problems.

11. What is a data source?

Answer: The place from which Power BI obtains data, such as Excel, CSV, SQL Server, or other supported sources.

12. What is Applied Steps in Power Query?

Answer: A list of transformations performed on the data. Each step records an operation such as filtering, renaming, or changing data types.

13. What is Merge Queries?

Answer: It combines columns from two tables by matching related values, similar to a database join.

14. What is Append Queries?

Answer: It stacks rows from two or more tables with compatible columns into one table.

3. Data Modeling

15. What is a data model?

Answer: A structured collection of tables, relationships, columns, and calculations used to analyze data.

16. What is a relationship?

Answer: A connection between tables based on related columns, such as CustomerID in a Customer table and Sales table.

17. What is a primary key?

Answer: A column that uniquely identifies each row in a table.

18. What is a foreign key?

Answer: A column that references a key in another table and helps establish relationships.

19. What is a star schema?

Answer: A model with a central fact table connected to surrounding dimension tables.

20. What is a fact table?

Answer: A table that usually stores measurable business events, such as sales, quantity, or revenue.

21. What is a dimension table?

Answer: A table that provides descriptive information used to analyze facts, such as customer, product, or date.

4. Visualizations

22. What is a visualization?

Answer: A graphical representation of data, such as a bar chart, line chart, table, card, or map.

23. What is a slicer?

Answer: An interactive filter that allows users to select values and change the data displayed in visuals.

24. What is a Card visual?

Answer: A visual commonly used to display a single important value, such as Total Sales or Total Customers.

25. When is a line chart useful?

Answer: It is useful for showing trends over time, such as monthly sales.

26. When is a bar/column chart useful?

Answer: It is useful for comparing values across categories, such as sales by product or region.

27. What is a KPI visual?

Answer: A visual designed to show a key performance indicator against a target or goal.

5. DAX Basics

28. What is DAX?

Answer: DAX (Data Analysis Expressions) is the formula language used in Power BI for calculations and data analysis.

29. What is a measure?

Answer: A dynamic calculation evaluated according to the current filter and report context, such as Total Sales.

30. What is a calculated column?

Answer: A column calculated row-by-row using a DAX expression and stored in the model.

31. Difference between measure and calculated column?

Answer: A measure is calculated when needed in the current context; a calculated column is calculated for each row and stored in the model.

32. What is SUM() in DAX?

Answer: SUM() adds the values in a numeric column.

33. What is CALCULATE()?

Answer: CALCULATE() evaluates an expression after modifying the filter context. It is one of the most important DAX functions.

34. What is filter context?

Answer: The set of filters that determines which data is considered when a measure is evaluated.

6. Power BI Interview / Viva

35. What is a workspace?

Answer: A collaborative area in Power BI Service where reports, dashboards, semantic models, and other content can be managed.

36. What is a semantic model?

Answer: The data model that provides tables, relationships, calculations, and metadata used by reports and other Power BI content.

37. What is refresh?

Answer: The process of updating the data in a semantic model from its connected data sources.

38. What is drill-down?

Answer: A feature that lets users move from a higher-level view to more detailed levels in a hierarchy.

39. What is drill-through?

Answer: A feature that opens a detailed report page filtered to the selected data point.

40. What is a Power BI Gateway?

Answer: A gateway enables Power BI Service to securely connect to certain on-premises data sources for data refresh or other supported operations.

■ Quick Revision Sheet

Concept	Remember This
Power Query	Clean & transform data
Data Model	Tables + relationships
Fact Table	Business events / numbers
Dimension Table	Descriptive information
Measure	Dynamic calculation
Calculated Column	Row-by-row stored calculation
DAX	Power BI calculation language
Slicer	Interactive filter
Dashboard	Single-page visual overview
Report	One or more pages of visuals
Refresh	Update data from source

■ Interview Tip

For a beginner Power BI interview, be ready to explain the complete flow: **Get Data** → **Power Query** → **Data Model** → **DAX** → **Visualizations** → **Report** → **Publish** → **Refresh**.

Practice these questions aloud and try to explain each answer in your own words.